Communication Systems Lecture-6: Pulse Code Modulation

Chadi Abou-Rjeily

Department of Electrical and Computer Engineering
Lebanese American University
chadi.abourjeily@lau.edu.lb

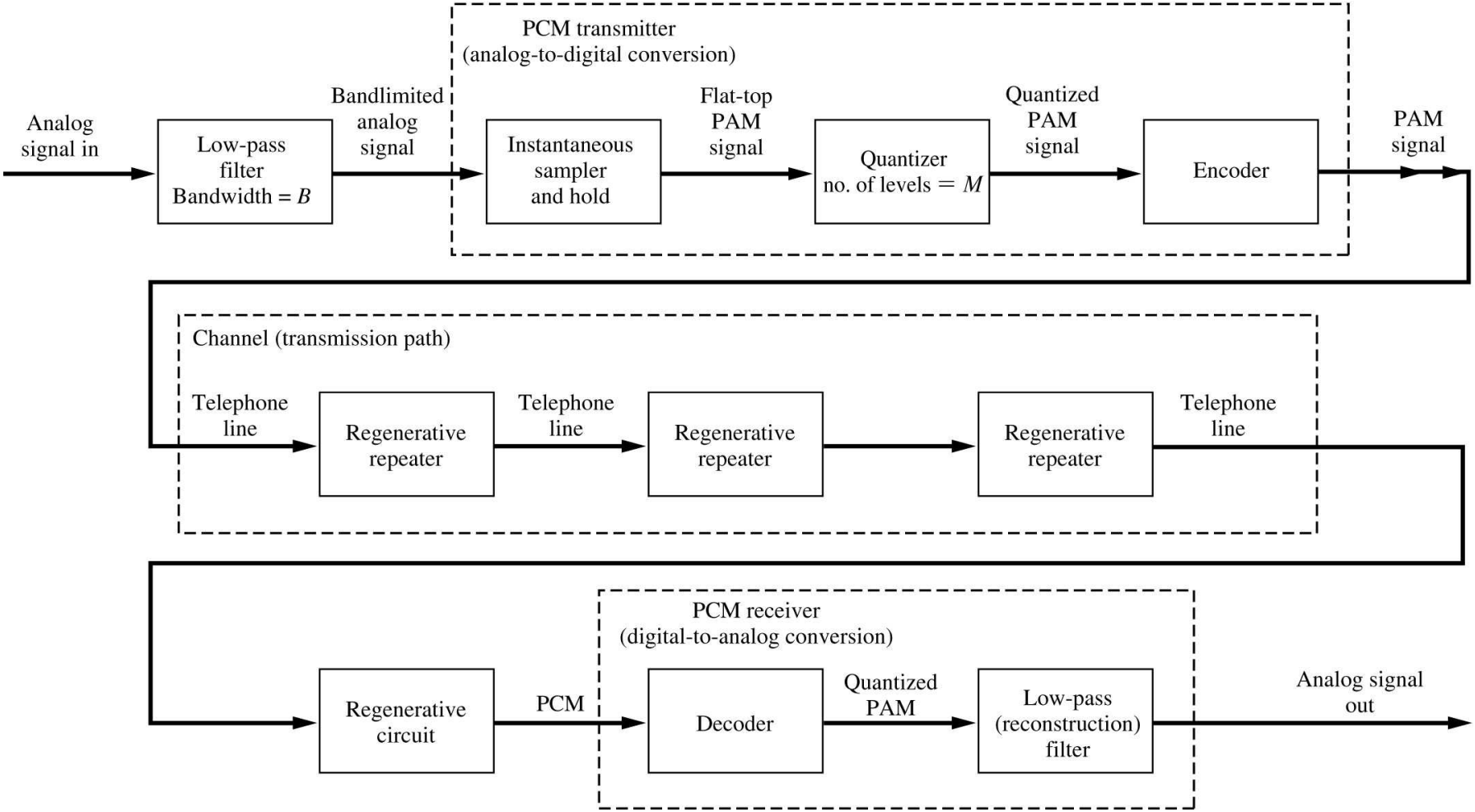
September 19, 2024

PCM (1)

Pulse Code Modulation (PCM): is essentially Analog-to-Digital conversion (A/D) of a special type where the information contained in the instantaneous samples of an analog signal is represented by digital words in a serial bit stream.

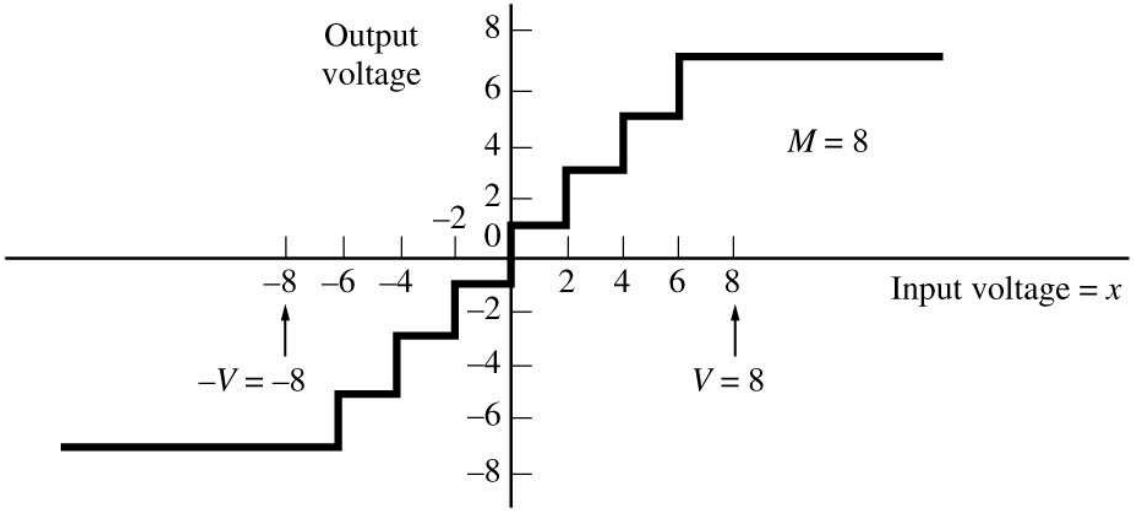
- The PCM signal is generated by carrying out three basic operations:
  - Sampling: where the analog signal is transformed into a flat-top PAM signal. In other words, the analog signal is represented by a certain number of samples.
  - Quantization: the samples of the analog signal can take an infinite number of values. The signal at the output of the quantizer can take M possible values. In this case, each sample is replaced by the closest level (among the M possible levels).
  - Encoding: each level is mapped into a sequence of n bits. The output of the decoder is a binary codeword having a length n where $M = 2^n$.

PCM (2)

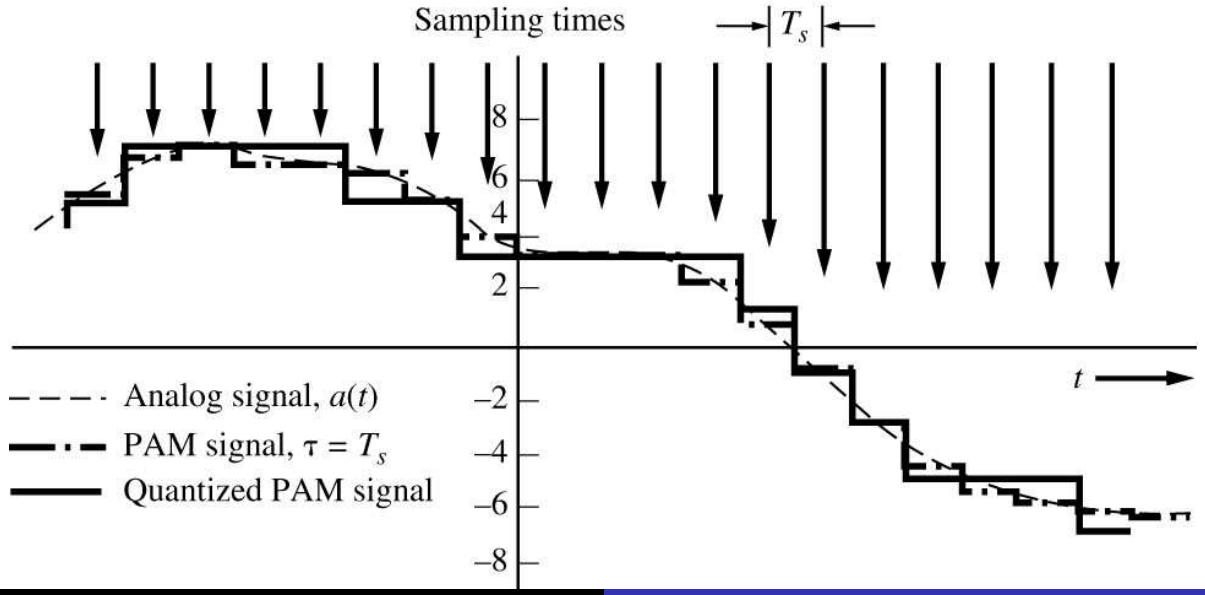

PCM transmitter
(analog-to-digital conversion)
Analog signal in
Low-pass filter Bandwidth = B
Bandlimited analog signal
Instantaneous sampler and hold
Flat-top PAM signal
Quantizer no. of levels = M
Quantized PAM signal
Encoder
PAM signal
Channel (transmission path)
Telephone line
Regenerative repeater
Telephone line
Regenerative repeater
Regenerative repeater
Telephone line
PCM receiver
(digital-to-analog conversion)
Regenerative circuit
PCM
Decoder
Quantized PAM
Low-pass (reconstruction) filter
Analog signal out

PCM (3)

- In the next figure $M = 8$:



(a) Quantizer Output-Input Characteristics



PCM (4)

- It is up to the system designer to specify the exact codeword that will represent a particular quantized level.
- The next example shows a Gray code with $M = 8$ ($n = 3$ bits):

Quantized voltage	Gray codeword
+7	110
+5	111
+3	101
+1	100
-1	000
-3	001
-5	011
-7	010

Mirror image except for the sign bit

- For the Gray codes:
  - The first bit is the sign bit.
  - Consecutive levels differ by only one bit. Consequently, noise that moves the signal sample from one quantization level to an adjacent level results in a single bit error in the received PCM codeword.

Bandwidth of PCM signals (1)

- Due to the nonlinear operations introduced by PCM, the spectrum of PCM signals is not directly related to the spectrum of the input analog signal (as with PAM).
- The bandwidth depends on:
  1. Bit rate.
  2. Waveform pulse shape.
- Since a sample is generated every T_s seconds and since every sample is encoded into n bits, the bit rate of PCM systems is:

$$R = \frac{n}{T_s} = nf_s$$

- For no aliasing, it is required that $f_s \geq 2B$ resulting in $R \geq 2nB$.

Bandwidth of PCM signals (2)

- It will be shown that the bandwidth of the binary encoded PCM waveform is bounded by:

$$B_{\mathrm{PCM}} \geq \frac{1}{2}R = \frac{1}{2}nf_s$$

  - The minimum bandwidth of $\frac{1}{2}R$ is obtained only when sinc-type pulses are used.
  - Usually, rectangular-type pulses are used. In this case, the bandwidth of the PCM waveform will be larger than this minimum.

Effects of noise (1)

- Two main effects produce noise in PCM systems:
  - Quantizing noise that results from the M-level approximation of the samples.
    - This can be reduced by increasing the binary length of each codeword (in other words by increasing $M = 2^n$).
  - Channel noise that results in bit errors in the recovered PCM signal.
- It can be shown that the ratio of the average power of the recovered signal to the average noise power is:

$$\left(\frac{S}{N}\right)_{out} = \frac{M^2}{1 + 4(M^2 - 1)P_e}$$

where P_e is the probability of bit error in the recovered binary PCM signal at the receiver before it is converted back into an analog signal.

  - Note that the above equation is valid under the assumption that the signal has a uniform distribution between $-V$ and V where V can take any positive value.

Effects of noise (2)

- To consider the quantizing noise alone, we set $P_e = 0$ resulting in:

$$\left(\frac{S}{N}\right)_{out} = M^2$$

- In decibels:

$$\left(\frac{S}{N}\right)_{out,dB} = 10\log_{10}(M^2) = 10\log_{10}(2^{2n})$$
$$= 10\log_{10}(4^n) \approx 6.02n$$

- This is called the 6-dB rule where a 6-dB improvement in the SNR is obtained when an additional bit is added to each PCM codeword.

Example: PCM for telephone systems

- An analog audio telephone signal occupies a band from 300 Hz to 3.4 kHz.
- The minimum sampling rate is $f_s = 2 \times 3.4 = 6.8$ kHz.
- In order to allow the use of practical low-pass filters with reasonable transition bands, the signal is oversampled with a sampling frequency of 8 ksamples/s.
- Each sample is represented by 8 bits and the data rate is:

$$\begin{aligned} R &= nf_s \\ &= [8\ (\text{bits/sample})\ ] \times [8\ (\text{ksamples/s})\ ] = 64\ (\text{kbits/s}) \end{aligned}$$

- In this case, the minimum absolute bandwidth is:

$$B_{min} = \frac{1}{2}R = 32\ \ \text{kHz}$$

- The average signal-to-quantizing-noise ratio is:

$$\left(\frac{S}{N}\right)_{out,dB} \approx 6n = 48\ \text{dB}$$

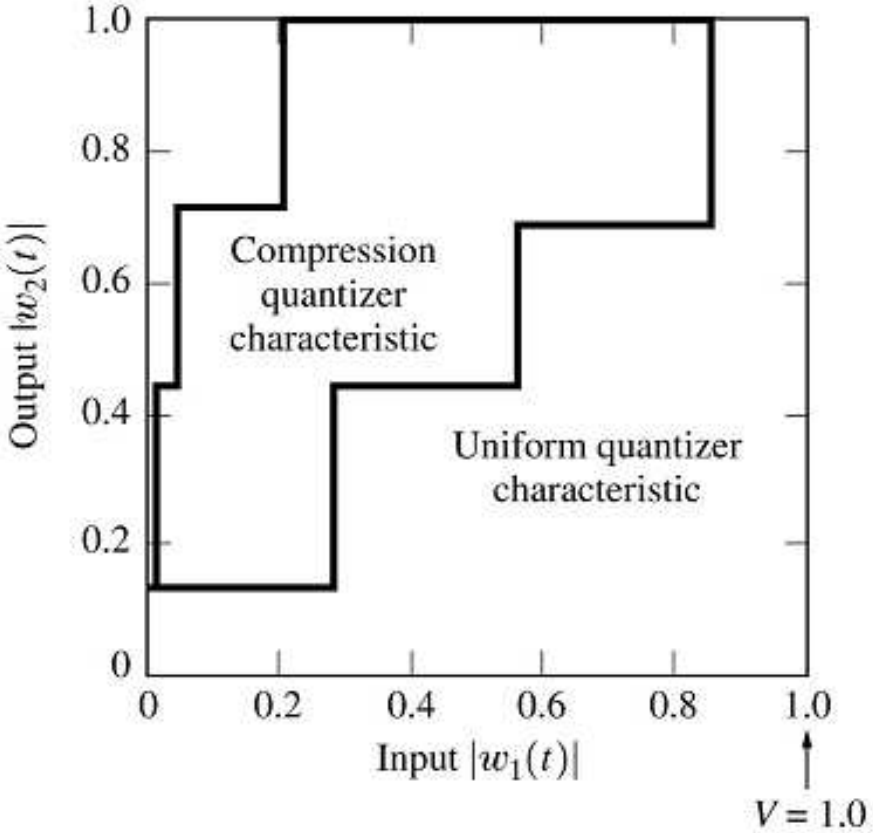
Nonuniform Quantizing (1)

Uniform quantization presents the following limitations:

- Voice analog signals are more likely to have amplitude values near zero than at the extreme peak values allowed.
- When uniform quantizing is used, the level of quantizing noise is independent of the signal amplitude. Consequently, small amplitudes suffer more from quantizing effects with respect to large amplitudes.

Nonuniform Quantizing (2)

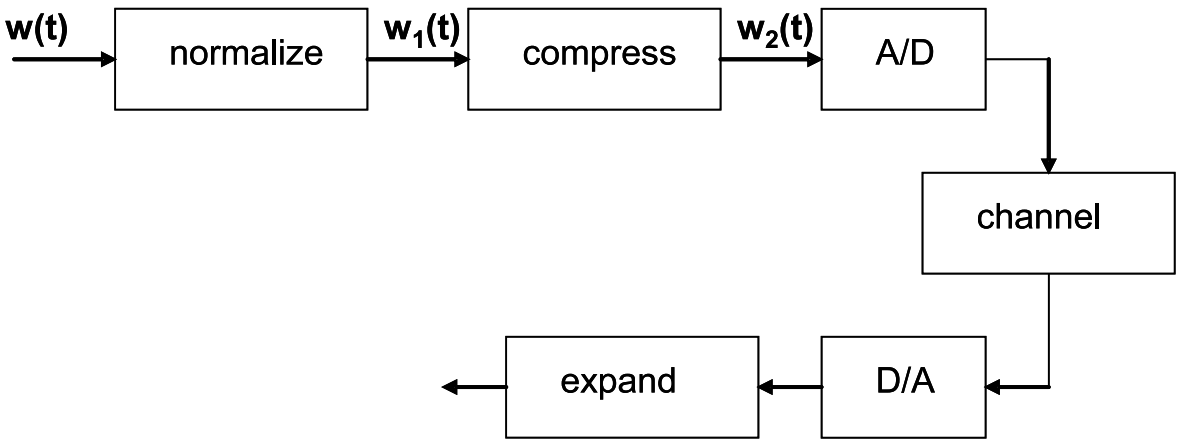
- One way for solving this problem is by applying nonuniform quantization. In this case:
  - Quantizing steps are made small for small amplitude levels.
  - Quantizing steps are made large for large amplitude levels.
  - This quantization can be achieved by the use of a variable step size.



(a) $M = 8$ Quantizer Characteristic

Nonuniform Quantizing (3)

- The effect of nonuniform quantization can be achieved by:
  - First passing the analog signal through a compression (nonlinear) amplifier.
  - Then applying uniform quantization.
- At the receiver side, the inverse operation (expanding) must be performed.



$$w_1(t) = \frac{w(t)}{\max|w(t)|} \quad \Rightarrow \quad |w_1(t)| \leq 1$$

Nonuniform Quantizing (4)

μ-law: The μ-law type of compression is defined by:

$$|w_2(t)| = \frac{\ln\left(1 + \mu\,|w_1(t)|\right)}{\ln\left(1 + \mu\right)}$$

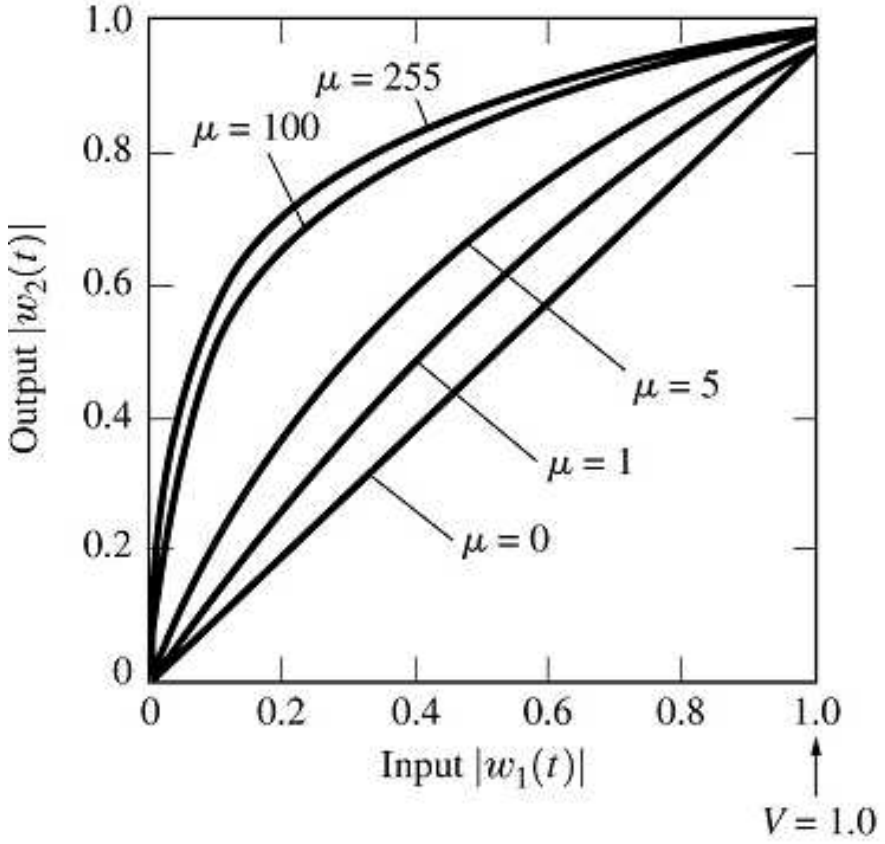
- In the United States, Canada and Japan, the telephone companies use a $\mu = 255$ compression characteristic in their PCM systems.

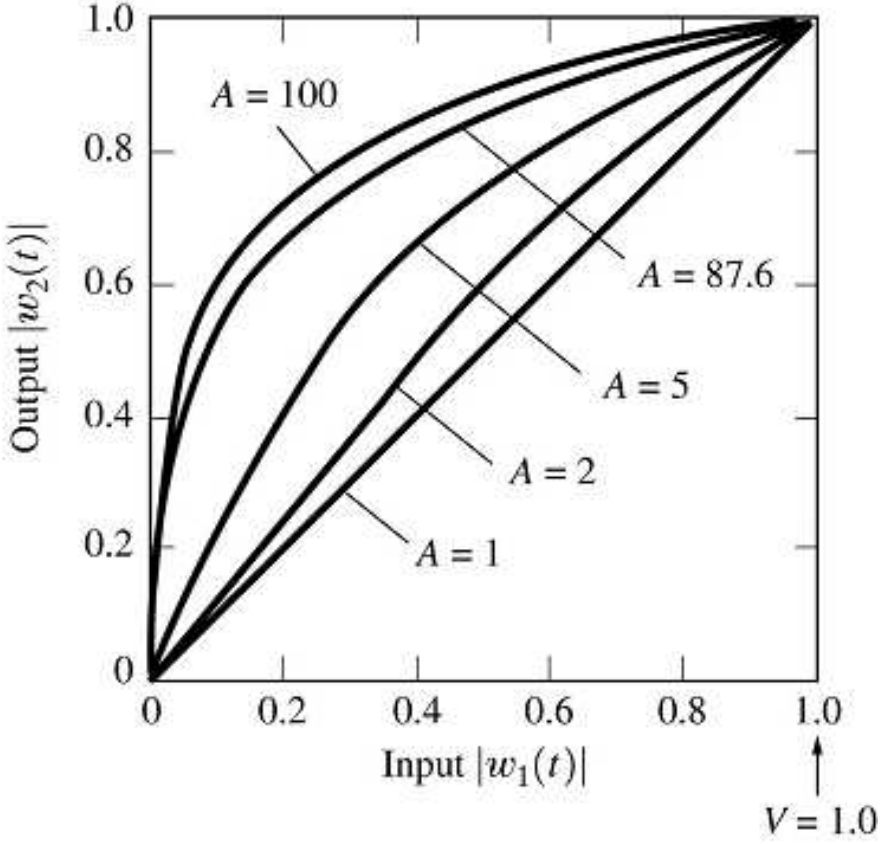
A-law: The A-law type of compression is defined by:

$$|w_2(t)| = \begin{cases} \frac{A|w_1(t)|}{1+\ln A}, & 0 \leq |w_1(t)| \leq \frac{1}{A}; \\ \frac{1+\ln(A|w_1(t)|)}{1+\ln A}, & \frac{1}{A} \leq |w_1(t)| \leq 1. \end{cases}$$

- The A-law is used mainly in Europe. A typical value of A is $A = 87.6$.

Nonuniform Quantizing (5)



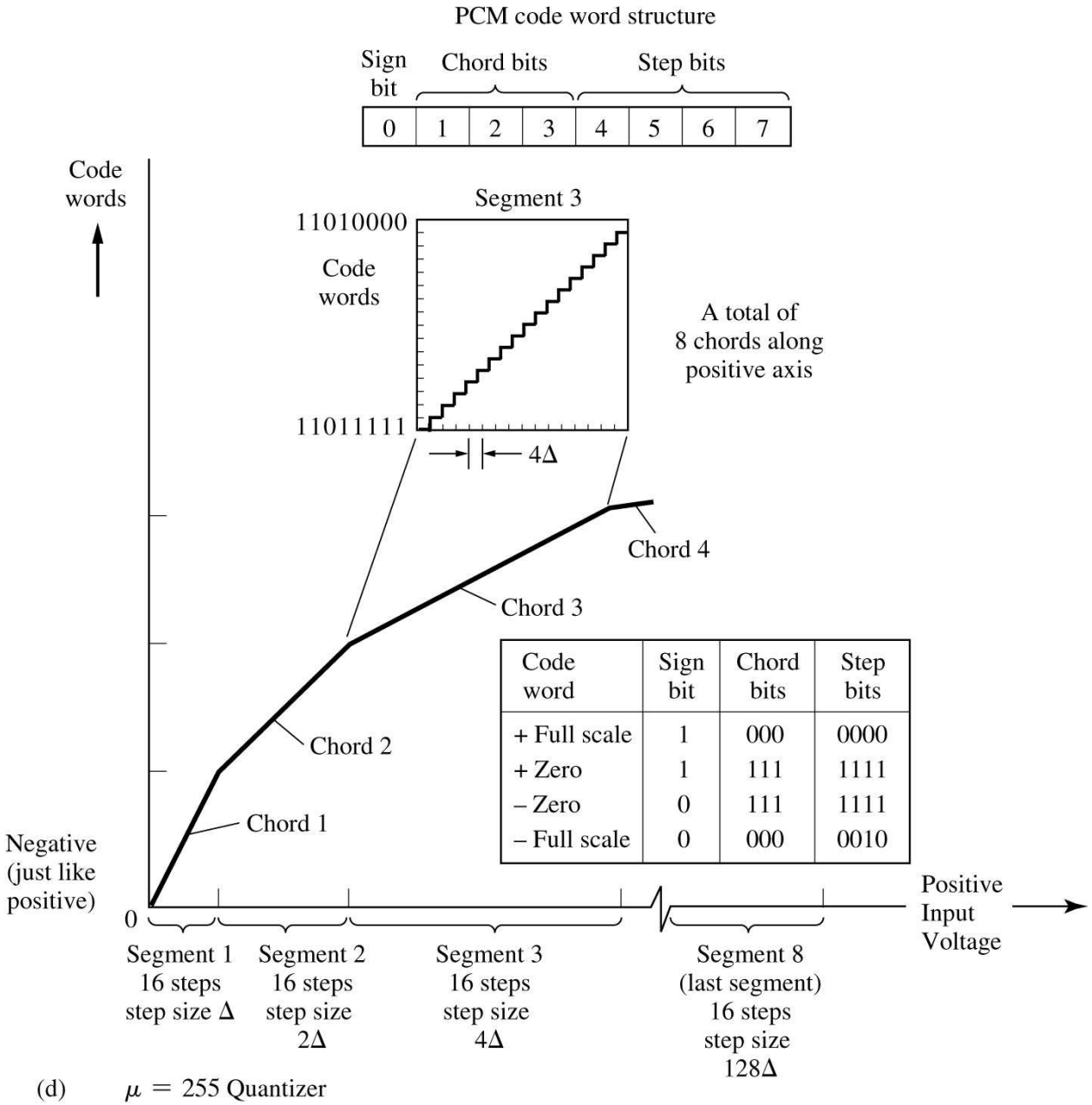


Nonuniform Quantizing (6)

- In practice, the nonlinear characteristics of the μ-law and A-law are approximated by piecewise linear chords.
- For example, the telephone companies in the United states use the $\mu = 255$ compression for generating 8-bit PCM codewords. For this 8-bit quantization:
  - 8 chords are used to approximate the μ-law compression curve for the positive portion of the signal.
  - Each chord is approximated by a uniform quantizer with 16 steps.
  - The length of the steps depend on the chord number. For example, chord 1 has 16 steps having a length of Δ each. Chord 2 has 16 steps having a length of 2Δ each. In general, the step size in the n-th chord is $2^{n-1}\Delta$.
  - The same procedure is applied on the negative part of the signal.
- Consequently, the total number of steps is:

$$N_{steps} = 2 \times 8 \times 16 = 256$$

Nonuniform Quantizing (7)



Code word	Sign bit	Chord bits	Step bits
+ Full scale	1	000	0000
+ Zero	1	111	1111
– Zero	0	111	1111
– Full scale	0	000	0010

(d) $\mu = 255$ Quantizer

Nonuniform Quantizing (8)

- As can be seen from the previous figure, the 8-bit PCM codeword consists of:
  - 1 sign bit that denotes a positive or negative input voltage.
  - 3 chord bits that denote the segment number.
  - 4 bits that denote the particular step within a segment.
- A segmenting technique similar to that used with the μ-law is applied with the A-law.
- The total length of the segments in the positive part of the signal is given by:

$$16\left[1+2+4+\cdots+128\right]\Delta = 16[255]\Delta = 4080\Delta$$

- Suppose that the analog signal is limited between $-V$ and V. The step size Δ is chosen to verify:

$$\Delta = \frac{V}{4080}$$

Nonuniform Quantizing (9)

Once again, it can be shown the output SNR follows the 6-dB rule:

$$\left(\frac{S}{N}\right)_{dB} = 6.02n + \alpha$$

where n is the number of bits per PCM codeword.

- For uniform quantization:

$$\alpha = 4.77 - 20\log_{10}\left(\frac{V}{x_{rms}}\right)$$

  where:
  - V is the peak value of the input analog signal.
  - x_{rms} is the root-mean-square value of the input analog signal.
- For the μ-law companding with <u>sufficiently large input levels</u>:

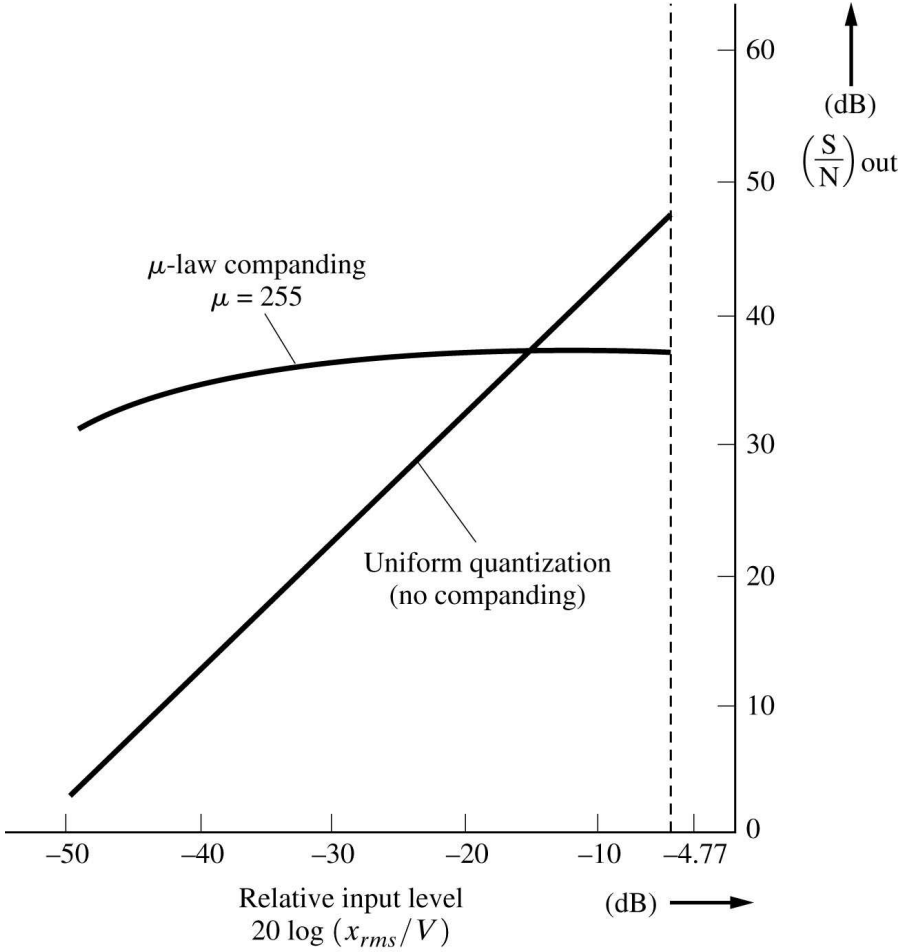
$$\alpha \approx 4.77 - 20\log_{10}\left[\ln(1+\mu)\right]$$

- For the A-law companding with <u>sufficiently large input levels</u>:

$$\alpha \approx 4.77 - 20\log_{10}\left[1+\ln(A)\right]$$

Nonuniform Quantizing (10)

- The ratio V/x_{rms} is called the loading factor.
- Notice that the output SNR is a function of the input level for the case of uniform quantizing, but it is relatively insensitive to the input level for μ-law and A-law companding.



Nonuniform Quantizing (11)

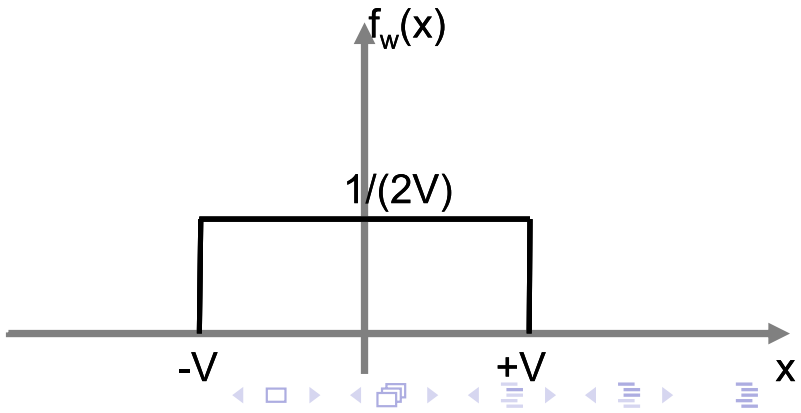
For uniform quantizing with a uniformly distributed input signal, $\alpha = 0$ and the output SNR takes the simple value given at the beginning of this lecture:

$$\left(\frac{S}{N}\right)_{dB} = 6.02n$$

Proof:

- Consider the signal $w(t)$ that is uniformly distributed between $-V$ and V.
- The probability density function of $w(t)$ is given by:

$$f_w(x) = \begin{cases} \frac{1}{2V}, & -V \leq x \leq V; \\ 0, & \text{elsewhere.} \end{cases}$$



Nonuniform Quantizing (12)

- The mean value of $w(t)$ is given by:

$$\bar{x} = \int_{-\infty}^{+\infty} x f_w(x) \mathrm{d}x$$
$$= \frac{1}{2V} \int_{-V}^{+V} x \mathrm{d}x = 0$$

- The rms value can be calculated from:

$$x_{rms}^2 = \int_{-\infty}^{+\infty} (x - \bar{x})^2 f_w(x) \mathrm{d}x$$
$$= \frac{1}{2V} \int_{-V}^{+V} x^2 \mathrm{d}x = \cdots = \frac{V^2}{3}$$

- Consequently, for uniform quantization:

$$\alpha = 4.77 - 20 \log_{10} \left(\frac{V}{x_{rms}} \right)$$
$$= 4.77 - 20 \log_{10} \sqrt{3} = 0$$

Shannon's Theorem

- The capacity of a channel is given by (in bits/s):

$$C = B \log_2 \left(1 + \frac{S}{N}\right)$$

  where:
  - B is the bandwidth of the channel.
  - $\frac{S}{N}$ is the signal-to-noise ratio.
- **Shannon's theorem**: a data rate R can be achieved over a communication channel with an arbitrarily small probability of error (P_e) if and only if:

$$R \leq C$$

Example: V.90 56-kb/s PCM computer modem (1)

- This modem employs μ-law PCM with $\mu = 255$.
- The codewords have a length of 7 bits: 1 sign bit, 3 chord bits and 3 bits to specify the step in a chord.
- The modem's clock is synchronized to the PCM 8-ksamples/s clock of the telephone company.
- This results in a data rate of:

$$R = nf_s = [7\ (\text{bits/sample})\ ]\times[8\ (\text{ksamples/s})\ ] = 56\ (\text{kbits/s})$$

- The SNR required to achieve this rate with an arbitrarily small probability of error can be calculated from Shannon's theorem:

$$R \leq C = B\log_2\left(1+\frac{S}{N}\right)$$

$$\Rightarrow \quad \frac{S}{N} \geq 2^{R/B} - 1$$

Example: V.90 56-kb/s PCM computer modem (2)

- Assuming that the filtering circuits limit the useful bandwidth of the telephone line to $B = 3.3$ kHz:

$$\left(\frac{S}{N}\right)_{dB} \geq 10 \log_{10} \left[2^{56/3.3} - 1\right] = 51.1 \text{ dB}$$

- If the SNR is less than 51.1 dB, the data rate of 56 kb/s can not be achieved with $P_e \approx 0$.
- In this case, the modem will fall back to a lower speed, such as 33.3 kb/s, 28.8 kb/s or 24 kb/s.